

EXPERIMENT-28 Medical Diagnosis

Aim

To develop a simple Prolog-based medical diagnosis system using symptoms.

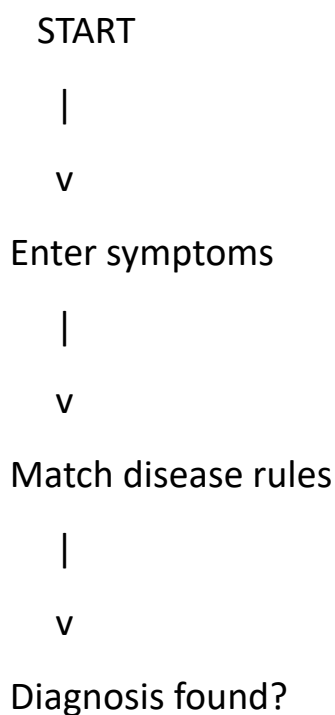
Objectives

- To represent symptoms as facts.
- To define disease rules.
- To infer possible diseases from symptoms.

Algorithm

1. Define symptoms.
2. Define rules connecting symptoms to diseases.
3. Enter symptoms as facts.
4. Query the diagnosis.
5. Display the possible disease.

Flowchart



/	\
Yes	No
v	v
Display	No diagnosis
diagnosis	found
\	/
STOP	

Program

% Medical Diagnosis System

symptom(fever).

symptom(cough).

symptom(body_pain).

symptom(rash).

symptom(fever).

symptom(sneezing).

symptom(runny_nose).

diagnosis(flu) :-

symptom(fever),

symptom(cough),

symptom(body_pain).

diagnosis(measles) :-

 symptom(fever),

 symptom(rash).

diagnosis(common_cold) :-

 symptom(sneezing),

 symptom(runny_nose).

Query

?- diagnosis(Disease).

Possible output:

Disease = flu ;

Disease = measles ;

Disease = common_cold ;

false.

Output

```
HelloWorld.pl x I/O AI Agent
% Diagnosis rules
diagnosis(flu) :-
    symptom(fever),
    symptom(cough),
    symptom(body_pain).

diagnosis(measles) :-
    symptom(fever),
    symptom(rash).

diagnosis(common_cold) :-
    symptom(sneezing),
    symptom(runny_nose).

% Main program
main :-
    write('Medical Diagnosis Results'), nl,
    write('-----'), nl,

    diagnosis(flu),
    write('Diagnosis: Flu'),
    nl,

    diagnosis(measles),
    write('Diagnosis: Measles'),
    nl,

    diagnosis(common_cold),
    write('Diagnosis: Common Cold').
```

STDIN
Input for the program (Optional)

Output:
Medical Diagnosis Results

Diagnosis: Flu
Diagnosis: Measles
Diagnosis: Common Cold

Result

The medical diagnosis system successfully identifies possible diseases based on symptoms.

Conclusion

Prolog can be used to create simple rule-based expert systems. This example is for academic purposes and is not a substitute for professional medical diagnosis.